East West University

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Project Report Task -1

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Submitted By-

Name	ID
Samia Hasan	2023-3-60-556
Shuvro Dev Sil	2023-3-60-224
Annur Tasnia	2023-3-60-429
Samia Naba	2023-3-60-578

Submitted To-

Dr.Raihun Ul Islam

Associate Professor

Department of Computer Science & Engineering

EMG Project Report for Task 1

Data Overview and Research Gap

1. Project overview

This project looks at surface electromyography (sEMG) signals recorded during different hand movements. The aim of the exploratory data analysis (EDA) is to understand the dataset, examine how the signals vary, and identify features that may help a machine-learning model distinguish the movements.

The six movement labels used in the notebook are cyl, hook, lat, palm, spher, and tip. The recordings contain two channels, called ch1 and ch2.

2. What the data looks like

Dataset item	Description
Databases	2 databases
Source files	8 MATLAB files
Movement classes	6 class : cyl, hook, lat, palm, spher, tip
Channels	2 EMG channels per trial
Total trials	2,700
Channel-level recordings	5,400 (two channels for each trial)
Class balance	450 trials per movement, or about 16.67% each
Database 1	5 files and 900 trials; 3,000 samples per channel
Database 2	3 files and 1,800 trials; 2,500 samples per channel

3. What we did in the EDA

- **Raw signal plots:** We plotted EMG signals over time to observe their patterns across movements and repeated trials.
- **Trial and source comparisons:** We compared repeated trials and looked at differences between recording sources. The source labels should not be assumed to represent unique participants without checking the documentation.
- **Feature extraction:** We used Python to calculate features such as RMS, MAV, variance, waveform length, zero-crossing rate, slope-sign-change rate, skewness, kurtosis, and frequency-related measurements.
- **Basic data checks:** The notebook reports no missing or infinite values in the feature table and reports no flat signals in its signal-quality checks.
- **Boxplots:** We compared the distributions of waveform length and zero-crossing rate across the six movements and two channels.
- **Correlation heatmap:** We checked which extracted features tend to vary together and which may contain overlapping information.
- **PCA:** The first two principal components explain about 52.19% of the feature variance combined. This summarizes the feature space; it is not a measure of classification accuracy.

4. Main findings so far

- The six movement classes are balanced, with 450 trials per class. This means the overall dataset does not contain more trials of one movement than another.
- The shown feature table has 2,700 rows and 36 columns. It includes extracted signal measurements along with metadata and the movement label.
- In the reported ANOVA ranking, rms_ch1 has the highest F-score (393.3), followed by mav_ch2 (377.5) and minimum_ch1 (354.5). These scores indicate differences in feature means across movement classes, but do not directly predict model accuracy.
- The boxplots show that waveform length and zero-crossing rate vary across movements. Some distributions overlap, so these features may not separate every movement on their own.
- The correlation heatmap shows strong relationships among several amplitude-related features, including RMS, variance, MAV, and standard deviation. This may indicate that some features provide similar information.
- The first two PCA components explain about 52.19% of the variance combined. This is useful for summarizing the data, but does not prove that the movements can be classified accurately.

5. Research gap

The literature review discusses feature extraction, combining time- and frequency-domain features, recognition across different users, session-to-session changes, and evaluation on data that differs from the training data. Based on those themes and the current EDA, the following questions are relevant to this project:

- **Generalization to new sources, people, or sessions:** A model may perform well on trials similar to its training data but perform worse on a new recording source, person, or session. The current plots show variation, but model generalization has not yet been tested.
- **Fair model evaluation:** If trials from the same source are randomly split between training and testing, related recordings may appear in both sets. A grouped split by source, person, or session—where those identities are confirmed—would give a more realistic test.
- **Which feature group works best?:** The EDA explores time-domain and frequency-domain features, but it does not establish whether time-domain features, frequency-domain features, or a combination gives the best classification performance.
- **Cross-database evaluation:** The dataset contains two databases with different trial counts and sample lengths. If their recording conditions and sampling rates are compatible, testing across databases could help assess generalization.
- **Clarifying source identities:** Labels such as female_1, male_1, and male_day_1 are source identifiers. The dataset documentation should be checked before treating them as separate people or sessions.

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7. What we can and cannot claim yet

Supported by current EDA	Not established yet
The dataset has 2,700 trials, six balanced classes, two channels, and eight source files.	That a model accurately recognizes all six movements.
Some feature distributions differ across movements.	That one feature is definitively the best feature.
Several extracted features are correlated.	That correlated features must always be removed.
The plots show variation across trials and sources.	That every source label represents a unique participant.
The literature review motivates generalization and session robustness.	That this project has already solved cross-person or cross-session generalization.

8. Conclusion

The EDA helped us understand the structure of the EMG dataset and how the signals and extracted features vary across six hand movements. The boxplots and correlation heatmap show that some features differ between movements, while other features are strongly related to each other. These findings help identify features worth testing, but they do not tell us the final classification accuracy.

The next step is to train and evaluate machine-learning models using a fair data-splitting strategy, ideally testing on recording sources, people, sessions, or databases that were not used during training. This will help determine how reliably the model can recognize hand movements beyond the data it has already seen.